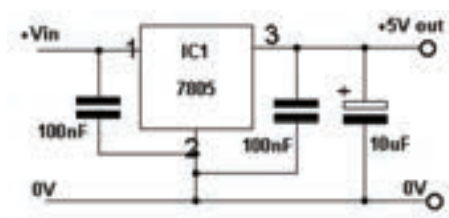


$$5.1\text{V} \times 0.101\text{A} = 0.52\text{ W}$$

For this application, you could probably get away with a $\frac{1}{2}$ W rated zener diode along with the 1 W rated 68 ohm resistor. Working with zener diodes to make power supplies is quite easy, but there is a tremendous price to pay in terms of lost power. In this example, the total power dissipated will be 1.2 W, with 58 percent of it being dissipated through the R resistor. This loss may be unacceptable in a battery-powered robot.

Linear voltage regulators

The zener diode regulator can be thought of as a tub of water with a hole at the bottom; the maximum pressure of the water squirting out is dependent on the level of water in the tub. Ideally, there should be more water coming into the tub than will be ever drawn to ensure that the pressure of the water coming out of the hole is constant. This means that a fair amount of water will spill over the edge of the tub. As was shown in the previous section, this is an extremely inefficient method of providing a regulated voltage, as the electrical equivalent of the water pouring over the edge is the power dissipated by R.



To improve upon the zener diode regulator's inefficiency, a voltage regulator that just lets out enough current at the regulated voltage is desired. The linear voltage regulator only allows the required current (at the desired voltage) out and works just like a car's

carburettor. In a carburettor, fuel is allowed to flow as required by the engine—if less is required than is available; a valve closes and reduces the amount of fuel that is passed to the engine. The linear voltage regulator works in an identical fashion: an output transistor in the regulator circuit only allows the required amount of current to the load circuit. The actual circuitry that implements the linear regulator is quite complex, but this is really not a concern of yours as it is usually contained within a single chip like the one shown in figure below.

The circuit shown here uses one of the most popular linear voltage regulators, the 7805 to regulate a high-voltage source to 5 V for digital electronic circuitry. Two of the most popular voltage regulators, the 7805 and